

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Trigonometric Functions

Week 3 Day 1 AM



SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

OUTLINE

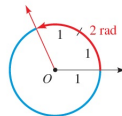
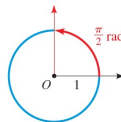
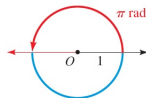
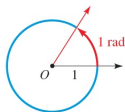
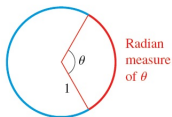
- ① Angle Measure
- ② Trigonometry of Right Triangles
- ③ Trigonometric Functions of Angles
- ④ Summary

RADIANS

You probably are familiar with the degree as a unit of measurement for angles. The degree is created by human based on the number of days the Earth takes to revolve around the sun and 360° is conveniently chosen because it's divisible by a lot of small primes. However, in calculus, the **default** (and natural) unit for angles is the **radian**. For instance, the derivative of $\sin x$ will not be $\cos x$ if x is not in radians.

DEFINITION:

If a circle of radius 1 is drawn with the vertex of an angle at its center, then the measure of this angle in **radians** (abbreviated **rad**) is the length of the arc that subtends the angle.



RELATIONSHIP BETWEEN DEGREES AND RADIANS

⚠ When no unit is given, the angle is assumed to be measured in radians.

RELATIONSHIP BETWEEN DEGREES AND RADIANS

$$180^\circ = \pi \text{ rad} \quad 1 \text{ rad} = \left(\frac{180}{\pi}\right)^\circ \quad 1^\circ = \frac{\pi}{180} \text{ rad}$$

1. To convert degrees to radians, multiply by $\frac{\pi}{180}$.
2. To convert radians to degrees, multiply by $\frac{180}{\pi}$.

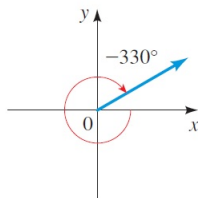
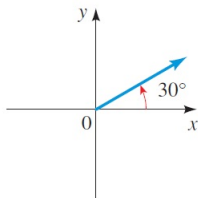
E.g.: $60^\circ = 60 \left(\frac{\pi}{180}\right) \text{ rad} = \frac{\pi}{3} \text{ rad}.$

E.g.: $\frac{\pi}{6} \text{ rad} = \frac{\pi}{6} \left(\frac{180}{\pi}\right)^\circ = 30^\circ.$

ANGLES IN STANDARD POSITION

An angle is in **standard position** if it is drawn in the xy -plane with its vertex at the origin and its initial side on the positive x -axis.

- If we turn **anti-clockwise** from the initial side, the angle created is **positive**.
- If we turn **clockwise** from the initial side, the angle created is **negative**.



LENGTH OF A CIRCULAR ARC

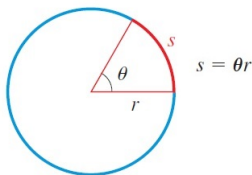
The circumference of a circle of radius r is $2\pi r$. An arc of this circle that subtends an angle θ has a length that is the fraction $\frac{\theta}{2\pi}$ of the circumference of the entire circle, i.e. the arc length ratio is the same as the angle ratio. So the length s of this arc is

$$s = \frac{\theta}{2\pi} \times 2\pi r = r\theta.$$

CIRCULAR ARC LENGTH FORMULA

In a circle of radius r the length s of an arc that subtends a central angle of θ radians is

$$s = r\theta.$$



AREA OF A CIRCULAR SECTOR

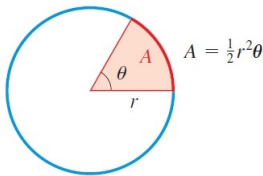
The area of a circle of radius r is $A = \pi r^2$. A sector of this circle with central angle θ has an area that is the fraction $\frac{\theta}{2\pi}$ of the area of the entire circle, i.e. the area ratio is the same as the angle ratio. So the area of this sector is

$$A = \frac{\theta}{2\pi} \times \pi r^2 = \frac{1}{2} r^2 \theta.$$

SECTOR AREA FORMULA

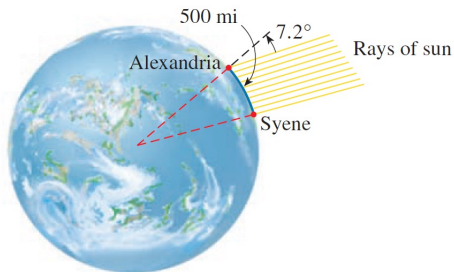
In a circle of radius r the area A of a sector with a central angle of θ radians is

$$A = \frac{1}{2} r^2 \theta.$$

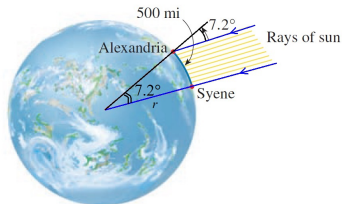


ACTIVITY 1

The Greek mathematician Eratosthenes (ca. 276–195 b.c.) measured the circumference of the earth from the following observations. He noticed that on a certain day the sun shone directly down a deep well in Syene (modern Aswan). At the same time in Alexandria, 500 miles north (on the same meridian), the rays of the sun shone at an angle of 7.2° to the zenith. Use this information and the figure to find the radius and circumference of the earth.



ACTIVITY 1: SOLUTION



Let the radius of the earth be r . From the given information, the angle subtended by the arc between Syene and Alexandria is 7.2° . Thus, we have

$$s = r\theta \Leftrightarrow 500 = r \times 7.2 \left(\frac{\pi}{180} \right) \Leftrightarrow r = \frac{500 \times 180}{7.2 \times \pi} \approx 3978.87 \text{ mi.}$$

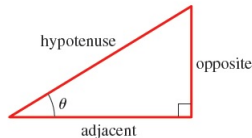
The modern-day measurement of the radius of the earth is somewhere between 3950 mi and 3963 mi (because the earth is not a perfect sphere). The circumference of the earth would be $2\pi r = 2\pi \times 3978.87 = 25000 \text{ mi.}$

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TRIGONOMETRIC RATIOS

Consider a right triangle with θ as one of its acute angles. The trigonometric ratios are defined as follows.



DEFINITION:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\begin{aligned} \csc \theta &= \frac{1}{\sin \theta} \\ &= \frac{\text{hypotenuse}}{\text{opposite}} \end{aligned}$$

$$\begin{aligned} \sec \theta &= \frac{1}{\cos \theta} \\ &= \frac{\text{hypotenuse}}{\text{adjacent}} \end{aligned}$$

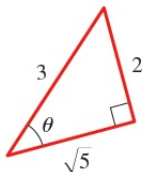
$$\begin{aligned} \cot \theta &= \frac{1}{\tan \theta} \\ &= \frac{\text{adjacent}}{\text{opposite}} \end{aligned}$$

The symbols we use for these ratios are abbreviations for their full names: **sine**, **cosine**, **tangent**, **cosecant**, **secant**, **cotangent**.

TRIGONOMETRIC RATIOS - EXAMPLE

Remark: The definition of the trigonometric ratios is independent of the size of the right triangle, because all right triangles with angle θ are similar, hence the ratios of any two sides are the same for all such triangles.

E.g.:



$$\sin \theta = \frac{2}{3}$$

$$\cos \theta = \frac{\sqrt{5}}{3}$$

$$\tan \theta = \frac{2}{\sqrt{5}}$$

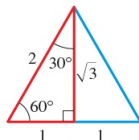
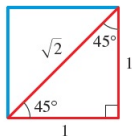
$$\csc \theta = \frac{3}{2}$$

$$\sec \theta = \frac{3}{\sqrt{5}}$$

$$\cot \theta = \frac{\sqrt{5}}{2}$$

SPECIAL TRIANGLES

You can get the two special triangles from a unit square and an equilateral triangle of side 2:

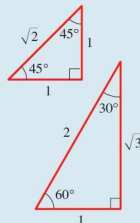


From these two special triangles, you can easily read off the special values of the trigonometric ratios:

SPECIAL VALUES OF THE TRIGONOMETRIC FUNCTIONS

The following values of the trigonometric functions are obtained from the special triangles.

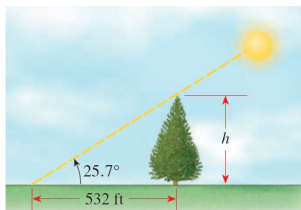
θ in degrees	θ in radians	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\csc \theta$	$\sec \theta$	$\cot \theta$
0	0	0	1	0	—	1	—
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	2	$\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	$\sqrt{2}$	$\sqrt{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2\sqrt{3}}{3}$	2	$\frac{\sqrt{3}}{3}$
90°	$\frac{\pi}{2}$	1	0	—	1	—	0



APPLICATIONS OF TRIGONOMETRIC RATIOS

Fun fact: Despite not having advanced tools and machinery, **astronomers** in ancient time used angles and trigonometry to estimate distances of heavenly bodies, such as the radius of the moon, the distance between the earth and the sun, etc.

E.g.: A giant redwood tree casts a shadow 532 ft long. Find the height of the tree if the angle of elevation of the sun is 25.7° .



If we let its height be h , then we see from the figure that

$$\frac{h}{532} = \tan 25.7^\circ \Leftrightarrow h = 532 \tan 25.7^\circ \approx 256 \text{ ft.}$$

ACTIVITY 2

Here is a way to estimate the distance between the earth and the moon: When the moon is seen at its zenith at a point A on the earth, it is observed to be at the horizon from point B (see the following figure). Points A and B are 6155 mi apart, and the radius of the earth is 3960 mi.

- (a) Find the angle θ in degrees.
- (b) Estimate the distance from point A to the moon.



ACTIVITY 2: SOLUTION

(a) By the circular arc length formula, we have

$$6155 = 3960\theta \Leftrightarrow \theta = \frac{6155}{3960} \text{ rad} = \frac{6155}{3960} \times \left(\frac{180}{\pi}\right)^\circ \approx 89.054425^\circ.$$

(b) Hence,

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\Leftrightarrow \text{hypotenuse} = \frac{\text{adjacent}}{\cos \theta}$$

$$= \frac{3960}{\cos 89.054425^\circ} \approx 239961 \text{ mi.}$$

Hence, the distance from A to the moon

= hypotenuse – earth's radius = $239961 - 3960 = 236001$ mi.

The modern-day measurement of the distance from the center of the earth to the center of the moon is 239000 mi.

OUTLINE

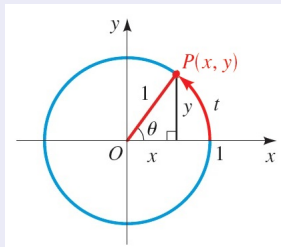
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TRIGONOMETRIC FUNCTIONS OVER $\mathbb{R}$

In the previous section, we defined the trigonometric ratios for acute angles only. Now we'll extend them to all angles in $\mathbb{R}$.

DEFINITION:

Consider the **unit** circle centered at the origin of the xy -plane. Let $P(x, y)$ be the point on the unit circle determined by an arc length t measured anti-clockwise from the point $(1, 0)$. Since the **radius is 1**, the arc subtends an angle $\theta = t$ rad ($\theta = -t$ rad if the arc length t is measured clockwise) at the center of the circle. With this, we define the **trigonometric**



functions

$$\sin \theta = y$$

$$\cos \theta = x$$

$$\tan \theta = \frac{y}{x} \quad (x \neq 0)$$

$$\csc \theta = \frac{1}{y} \quad (y \neq 0)$$

$$\sec \theta = \frac{1}{x} \quad (x \neq 0)$$

$$\cot \theta = \frac{x}{y} \quad (y \neq 0).$$

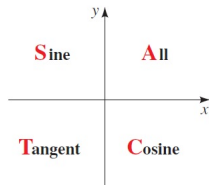
Note that we can turn a few rounds and get $t > 2\pi$, thus $\theta \in \mathbb{R}$.

SIGNS OF THE TRIGONOMETRIC FUNCTIONS

Remark: The choice of using a unit circle to define the trigonometric functions is merely to simplify things. In fact, we can use circles of any radius $r > 0$ to define them, in which case

$$\begin{aligned} \sin \theta &= \frac{y}{r} & \cos \theta &= \frac{x}{r} & \tan \theta &= \frac{y}{x} \quad (x \neq 0) \\ \csc \theta &= \frac{r}{y} \quad (y \neq 0) & \sec \theta &= \frac{r}{x} \quad (x \neq 0) & \cot \theta &= \frac{x}{y} \quad (y \neq 0). \end{aligned}$$

From the definitions, we deduce their signs easily:



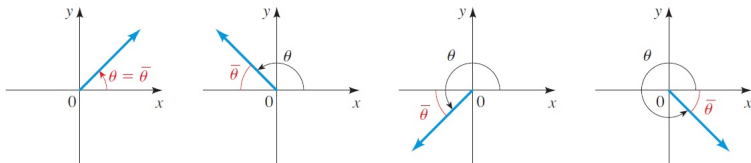
SIGNS OF THE TRIGONOMETRIC FUNCTIONS

Quadrant	Positive Functions	Negative Functions
I	all	none
II	sin, csc	cos, sec, tan, cot
III	tan, cot	sin, csc, cos, sec
IV	cos, sec	sin, csc, tan, cot

EVALUATING TRIGONOMETRIC FUNCTIONS FOR ANY ANGLE

To find the values of the trigonometric functions for any angle θ , you can follow these steps:

1. Find the **reference angle** $\bar{\theta}$ associated with the angle θ .
2. Determine the sign of the trigonometric function of θ by noting the quadrant in which θ lies.
3. The value of the trigonometric function of θ is the same, except possibly for sign, as the value of the trigonometric function of $\bar{\theta}$.

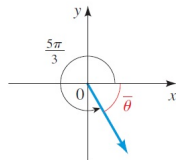


The reference angle $\bar{\theta}$ for an angle θ

EVALUATING TRIGONOMETRIC FUNCTIONS FOR ANY ANGLE - EXAMPLE

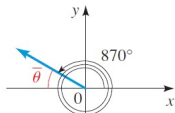
E.g.: To compute $\sin \frac{5\pi}{3}$, we first find the reference angle. Since $\frac{5\pi}{3}$ is in the 4th quadrant, its reference angle is $2\pi - \frac{5\pi}{3} = \frac{\pi}{3}$. We note that $\sin$ is negative in the 4th quadrant, thus

$$\sin \frac{5\pi}{3} = -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2}.$$



E.g.: To compute $\sin 870^\circ$, we first find the reference angle. After subtracting $2(360^\circ)$ from 870° , we get 150° , which is in the 2nd quadrant, so the reference angle is $180^\circ - 150^\circ = 30^\circ$. We note that $\sin$ is positive in the 2nd quadrant, thus

$$\sin 870^\circ = \sin 30^\circ = \frac{1}{2}.$$



TRIGONOMETRIC IDENTITIES

FUNDAMENTAL IDENTITIES

Reciprocal Identities

$$\csc \theta = \frac{1}{\sin \theta} \quad \sec \theta = \frac{1}{\cos \theta} \quad \cot \theta = \frac{1}{\tan \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1 \quad \tan^2 \theta + 1 = \sec^2 \theta \quad 1 + \cot^2 \theta = \csc^2 \theta$$

Notations: $\sin^2 \theta := (\sin \theta)^2$, $\cos^2 \theta := (\cos \theta)^2$,
 $\tan^2 \theta := (\tan \theta)^2$, $\sec^2 \theta := (\sec \theta)^2$, $\csc^2 \theta := (\csc \theta)^2$.

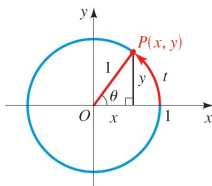
Remark: The last 3 identities are called the Pythagorean identities because they follow directly from Pythagoras' theorem. For e.g.,

$$\sin^2 \theta + \cos^2 \theta = y^2 + x^2 = 1.$$

$$(\div \cos^2 \theta) \quad \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$$

$$\Leftrightarrow$$

$$\tan^2 \theta + 1 = \sec^2 \theta.$$



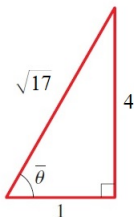
ACTIVITY 3

Given $\tan \theta = -4$ and θ is in the 2nd quadrant.

- (a) Let $\bar{\theta}$ be the reference angle of θ and draw a right triangle with an angle $\bar{\theta}$. What are the lengths of the 3 sides of the triangle?
- (b) By using the triangle and the sign of the trigonometric functions of θ , find the values of all other 5 trigonometric functions of θ .

ACTIVITY 3: SOLUTION

- (a) Since all reference angles are acute, we have $\tan \bar{\theta} = 4$ and so the ratio of the opposite side to the adjacent side is 4. By Pythagoras' theorem, the hypotenuse is $\sqrt{4^2 + 1^2} = \sqrt{17}$.



- (b) Since θ is in the 2nd quadrant, $\sin$ and $\csc$ are positive and the rest of the trigonometric ratios are negative:

$$\sin \theta = \frac{4}{\sqrt{17}}$$

$$\cos \theta = -\frac{1}{\sqrt{17}}$$

$$\tan \theta = -4$$

$$\csc \theta = \frac{\sqrt{17}}{4}$$

$$\sec \theta = -\sqrt{17}$$

$$\cot \theta = -\frac{1}{4}.$$

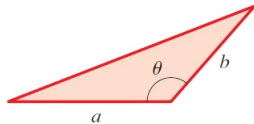
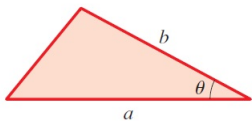
AREAS OF TRIANGLES

THEOREM

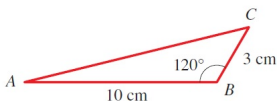
The area A of a triangle with sides of lengths a and b and with included angle θ is

$$A = \frac{1}{2}ab \sin \theta.$$

Please refer to page 497 – 498 of the textbook for the proof.



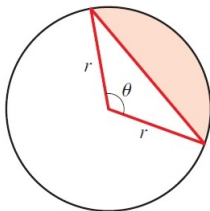
E.g.: The area of the following triangle is



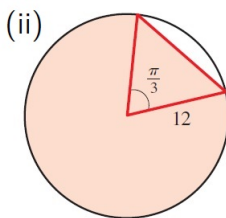
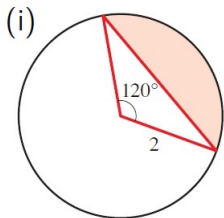
$$\begin{aligned} A &= \frac{1}{2}(10)(3) \sin 120^\circ = 15 \sin 60^\circ \\ &= 15 \frac{\sqrt{3}}{2} \approx 13 \text{ cm}^2 \end{aligned}$$

ACTIVITY 4

- (a) By using the **sector area formula**, derive the formula of the following shaded region of a circle in terms of r and θ .



- (b) Hence, find the areas of the following shaded regions.



ACTIVITY 4: SOLUTION

(a) The area of the shaded region is

$$\begin{aligned} A &= \text{area of the circular sector} - \text{area of the red triangle} \\ &= \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin \theta \\ &= \frac{1}{2}r^2(\theta - \sin \theta). \end{aligned}$$

(b) (i) $A = \frac{1}{2} \cdot 2^2 \left(120 \cdot \frac{\pi}{180} - \sin \frac{2\pi}{3} \right) = 2 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) \approx 2.45674.$

(ii) $A = \text{area of the shaded sector} + \text{area of the red triangle}$
$$\begin{aligned} &= \frac{1}{2}r^2 \left(2\pi - \frac{\pi}{3} \right) + \frac{1}{2}r^2 \sin \frac{\pi}{3} = \frac{1}{2}r^2 \cdot \frac{5\pi}{3} - \frac{1}{2}r^2 \sin \frac{5\pi}{3} \\ &= \frac{1}{2}r^2 \left(\frac{5\pi}{3} - \sin \frac{5\pi}{3} \right) = \frac{1}{2}(12)^2 \left(\frac{5\pi}{3} + \frac{\sqrt{3}}{2} \right) \\ &\approx 439.3449. \end{aligned}$$

We remark that the formula in part (a) still applies for $\theta > \pi$.

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SUMMARY

- Angle Measure
 - radians, relationship between degrees and radians.
 - angles in standard position.
 - length of a circular arc, area of a circular sector.
- Trigonometry of Right Triangles
 - trigonometric ratios.
 - special triangles.
 - applications of trigonometric ratios.
- Trigonometric Functions of Angles
 - trigonometric functions over $\mathbb{R}$.
 - signs of the trigonometric functions.
 - evaluating trigonometric functions for any angle.
 - trigonometric identities.
 - areas of triangles.

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 6.1, 6.2, 6.3.